Foundation of Embodied AI: Course Map

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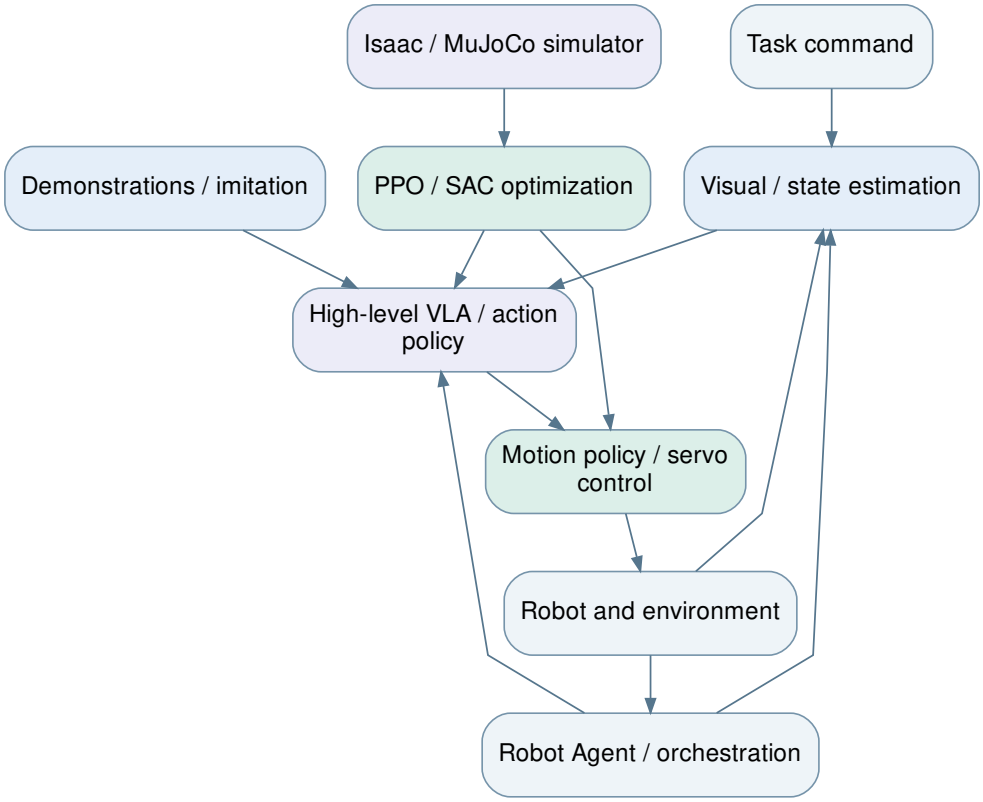
Fourteen topics are organized as tools → physics and perception → robot RL → demonstrations and generative policies → systems integration. Long-form derivations remain optional deep dives rather than strict prerequisites.

Six learning modules

Module	Topics	Outcome
0. Orientation	Systems map and guide	Explain perception–policy–control loops
1. Tools	Git, ROS 2	Manage code and understand ROS communication and TF
2. Physics and perception	Dynamics, calibration, vision foundation models, Isaac / MuJoCo	Relate physical models, sensors and simulation
3. Reinforcement learning	Motion RL, PPO, SAC	Design observations, actions, rewards and data collection
4. Demonstrations and models	Imitation Learning, Flow Matching, RL Post-training, GR00T / SONIC	Learn chunk policies from demonstrations and connect hierarchical controllers
5. Robot Agent integration	Robot Agent planning, tool use, verification and recovery	Orchestrate perception, policy and control for long-horizon tasks

Embodied AI system map



Reading paths

1. Beginners: orientation → essential Git and ROS 2 → dynamics → calibration → vision foundation models → simulation.
2. Motion control: dynamics → simulation → PPO / SAC → motion control RL.
3. VLA: calibration → vision foundation models → Imitation Learning → Flow Matching → RL Post-training → GR00T / SONIC → Robot Agent.
4. Mathematical deep dives: full PPO, SAC, and Flow Matching derivations are reference material; hands-on learning does not require deriving everything first.

Tutorial overview

1. Git Basics

Git Basics — Learn Git's working tree, staging area and commit history, then follow a GitHub workflow covering branches, pull requests, conflict resolution and common recovery commands.

2. ROS 2 Basics

ROS 2 Basics — Understand nodes, topics, services, actions, QoS and TF. Connect sensors, state estimation, policy inference and controllers through ROS 2 interfaces.

3. Camera Calibration

Camera Calibration — Study camera intrinsics, distortion, extrinsics and hand–eye calibration. Relate RGB-D observations and coordinate transforms to robot-frame object poses.

4. Vision Foundation Models

Vision Foundation Models — Compare segmentation, pose estimation, stereo depth and 3D foundation models. Follow four robotics workflows spanning known objects, stereo geometry, unknown shapes and human demonstrations.

5. Robot Dynamics

Robot Dynamics — Develop rigid-body motion and dynamics through coordinate frames, Jacobians, inertia and equations of motion. Connect contact, friction and control models to robot simulation.

6. Isaac / MuJoCo Simulation

Isaac / MuJoCo Simulation — Understand the roles of Isaac Sim, Isaac Lab and MuJoCo. Learn how robot assets, actuators, observations, rewards and environment interfaces form a simulation-based RL pipeline.

7. Motion Control RL

Motion Control RL — Connect MDP design and PPO/SAC to robot motion control. Study observations, action interfaces, rewards, resets, parallel simulation, domain randomization and Sim-to-Real considerations.

8. PPO

PPO — Derive PPO-Clip and Actor–Critic learning from policy gradients, Bellman targets and GAE. Explain Gaussian policy updates, gradient isolation and asymmetric robot observations.

9. SAC

SAC — Trace Q-learning, TD/SARSA and DQN toward entropy-regularized RL. Derive soft Bellman equations and modern SAC with twin critics, reparameterization and automatic entropy tuning.

10. Imitation Learning

Imitation Learning — Introduce BC, DAgger and ACT, then connect Diffusion Policy and Flow Matching to action-chunk generation. Explain temporal ensembling and inference-time versus training-time RTC.

11. Flow Matching

Flow Matching — Derive conditional Flow Matching, velocity fields and the training objective. Explain how ODE integration transforms noise into conditional robot action sequences.

12. RL Post-training

RL Post-training — Study RL refinement of pretrained Diffusion, Flow Matching and VLA policies. Compare generator updates with frozen-generator interfaces, Actor–Critic loops and real-robot feedback.

13. GR00T & SONIC

GR00T & SONIC — Use GR00T and SONIC to explain System 1/System 0 humanoid control. Connect high-level action chunks, motion representations, whole-body policies and low-level execution.

14. Robot Agents

Robot Agents — Distinguish foundation models, coding agents and agent harnesses. Study model-driven workflows, RAX exploration and compilation, ASPIRE trace repair, skill memory and task-level verification.